Red-Team Audit of the Geometric-Axis Hardness Claims (final_outputs2)

Scope. Independent recomputation and adversarial stress-testing of every quantitative claim in final_outputs2 (experiments I1--I7, J1--J6): the sd_along/s_along/s_orth/vel_entropy hardness predictors on MLAAD, the ASVspoof 2021 prospective predictions, the In-The-Wild transfer and per-speaker results, the AASIST replications, the axis-rotation law, the fusion/adaptation results, and the causal intervention battery. All audits run from the cached raw artifacts (embeddings, logits, scores) -- none rely on the original analysis code paths. Ten audits; code in experiments/axis_audits/, outputs in experiments/axis_audits/audits_outputs/audit{1..10}_*/.

1. What is proven beyond reasonable scrutiny

P1. sd_along -> MLAAD hardness (the core law). Independently recomputed: LOSO R^2 = 0.277 (claimed 0.273), permutation p = 0.0002, Spearman rho = +0.597 (p < 1e-6). It is the only test among all 83 recorded in the project that survives global BH-FDR across everything ever tested (q < 0.001). It survives: leave-one-system jackknife (0/61 folds lose significance), removal of the 5 most influential systems (rho rises to 0.66), partialling of n_utts/RMS/language/vel_entropy, estimator substitutions (ridge alpha, IQR spread, log-hardness), all three detector seeds, and an explicit winner's-curse simulation (wins 76.7% of discovery halves; 99.2% confirmation rate at p<0.05). Raising the min-utterance threshold strengthens it (rho = 0.60->0.64->0.69 at >=8/12/16 utts) -- the attenuation signature of a true effect measured noisily. [Audits 1, 2, 9]

P2. The hardness metric itself is sound. Split-half reliability 0.86; cross-seed agreement 0.89--0.97; shared-bona-pool dependence negligible (rank stability ~ 1.0 under bona bootstrap). The R^2 ceiling is ~0.86, so sd_along captures ~32% of explainable variance. [Audit 9]

P3. Axis rotation across recording domains is real -- and stronger than claimed. Each corpus's internal axis is estimated almost noiselessly (split-half cos 0.93--0.98), yet cross-corpus cosines are at/near the 768-d chance floor (E|cos| = 0.029): MLAAD<->ITW = 0.03--0.10 depending on frame, and MLAAD<->ASVspoof21 is actually negative (-0.14 to -0.21), consistent with the transferred-axis predictor P4 failing with the wrong sign. Corpus-internal axis estimation is therefore necessary, exactly as claimed. [Audit 4]

P4. Cross-detector agreement is domain-conditional. Verified matrix, plus a finding the paper missed: there are two same-domain cross-architecture pairs (AASIST-FT~WavLM-GAT = 0.553; AASIST-ZS~RobustGoat = 0.544, both ASVspoof-trained), and both sit above all four cross-domain pairs (0.30--0.36), including the same-architecture cross-domain pair (0.308). Caveat: no individual same-vs-cross difference is significant at n=61 (bootstrap p = 0.06--0.22); claim the pattern, not a significant gap. [Audit 7]

P5. I7 MLAAD axis fusion (0.272 -> 0.163) is clean. System-disjoint folds, lambda from train folds, unsupervised centroid axis; DeltaEER negative in all 5 folds x 3 seeds; fusion beats both components (axis alone 0.187). "Zero-training-cost" is fair for this experiment specifically. [Audit 5]

P6. The ungated-hook artifact (E9 correction) and the direction-content asymmetry. Gated-vs-ungated is seed-consistent (+0.010, all seeds, seed-level p = 0.04); artifact share recomputed at 81% ("75--80%" claim [ok]). sub_top (-0.032) vs sub_res (+0.004) is sign-consistent across all seeds with a 10x effect separation -- the directional-not-scalar mechanism story is qualitatively supported. [Audit 8]

2. What is suggestive but NOT proven

S1. ASVspoof 2021 prospective prediction (P3, rho = 0.60/0.64). The registered primary predictor P1 failed; P3 is a promoted secondary among 4 registered predictors; the earlier prospective attempt (J2) failed 9/9 tests; n = 13 attacks. Exact permutation p = 0.034/0.020 uncorrected -> Holm 0.13/0.08 within-family, BH q = 0.134; nothing survives. The "pre-registration" was written 63 s before scoring by the same script (no external commitment). In its favor: the two detector replications are quasi-independent (their hardness profiles correlate only rho = 0.21), measurement-noise bootstrap never crosses zero, and leave-one-attack-out stays in [0.49, 0.75]. Honest framing: a promising exploratory result requiring one genuinely held-out confirmation (e.g., ASVspoof21 DF, or the codec conditions) before any "prospective/pre-registered" language is used. [Audit 3]

S2. ITW per-speaker geometry (s_orth rho = +0.534). Survives the 7-test family correction (Holm p = 0.017) and confound partials -- but plain radius-of-gyration (rog12, rho = -0.561) is stronger, unreported, collinear with s_orth (rho = -0.687), and neither survives partialling the other. The robust claim is "a single spread/compactness factor predicts ITW speaker hardness"; the specific axis interpretation (off-axis residual) is not separable on current evidence, and a layer-0 velocity feature also predicting hardness hints at a channel confound. n = 29 speakers (not 58). [Audit 6]

S3. sd_along on AASIST-FT (rho = +0.349, p = 0.006). The Spearman form replicates and survives within-family BH, but the LOSO-R^2 form of the law fails on AASIST-FT (r^2 = -0.03, perm p = 0.13) and this is unreported; the paper switches metrics silently between detectors. Claim "rank-order replication," not "the law replicates." [Audits 1, 10]

S4. The I1 fine-grained causal effects. With seed as the inference unit (n = 3), no contrast survives BH; the utterance-level `` p-values are pseudo-replicated. The "small residual causal C effect" (iso_0.7, p = 0.024) is contradicted* -- seeds disagree in sign. Keep the artifact correction and the sub_top/sub_res asymmetry (sign-consistent); drop or re-power everything else (3--5 more seeds would suffice for the headline contrasts). [Audit 8]

3. What is wrong and must be fixed before submission

1. "cos(w_MLAAD, w_ASVspoof21) ~ 0.36" is false (true value negative; the 0.362 is J1's within-MLAAD cos(w_mean, w_lda) transplanted). [Audit 4]

2. The ITW "fusion" 0.363->0.292 is mislabeled -- 0.292 is the axis alone; actual fusion is 0.312 (fusion hurts on ITW for WavLM-GAT). No committed script generates itw_fusion_test.json. [Audit 5]

3. The 26--33-point "zero-training-cost fusion" gains under domain shift are misattributed. The supervised eval-corpus LDA probe alone beats the fused score (ITW: 0.101 vs 0.161; MLAAD: 0.100 vs 0.116); fusing the shifted detector in makes it worse, and ~3--4 EER points of the ITW number come from bona-speaker leakage across folds (strict speaker-disjoint: 0.193). The actionable story is supervised lightweight adaptation in frozen-SSL space (consistent with J3), not zero-cost fusion. [Audit 5]

4. Numerical/labeling errors: table2's "+0.316" is the triplet R^2 presented as sd_along; "true gated C-effect -0.0036" is the wrong contrast (-0.0023); "58 speakers" -> 29 analyzed; "13 interventions" -> 22 conditions; Figure 1's left panel prints s_along's stats (R^2 = 0.020) under the sd_along scatter, and its middle panel mixes I4's rho = -0.714 with J4's P3 axis label. Tally across the report: 8 verified, 6 mislabeled, 5 misleading, 2 contradicted. [Audit 10]

5. The I2 feature battery has zero FDR survivors (best q = 0.078) and should not be presented as supporting evidence without that disclosure. [Audit 1]

4. Failure scenarios tested and excluded

- Winner's curse / garden of forking paths on the core law -- excluded by split-half discovery/confirmation simulation. [A1]

- sd_along

5. Recommended actions (priority order)

1. Fix the two contradicted numbers and six mislabels (sec.3); regenerate every report number programmatically from artifacts.

2. Refram
predictor (

Bottom line. The central scientific claim -- per-system spread along the corpus-internal natural<->synthetic axis in frozen WavLM-L12 space predicts detection hardness, the axis rotates across recording domains, and hardness is training-domain-conditional rather than architecture-conditional -- survives aggressive red-teaming and is publishable. The ASVspoof21 "prospective" framing, the domain-shift "zero-training fusion" framing, the residual causal-C claim, and roughly a dozen packaged numbers do not survive and would be found by competent reviewers; they should be corrected or reframed exactly as itemized above.